

Should You Take the Singles Line?

Mathematical problems often have multiple solutions, and the question “should you take the singles line” is a prime example.

One solution is trivial: “no, you shouldn’t take the singles line.”

This is the answer if you’re shepherding your mother-in-law around the mountain, introducing your never-skied-before girlfriend to the slopes, or corralling a gaggle of erratic and inattentive four-year-olds for the day. Ditching them would be bad, even if you regret volunteering for the assignment. “No” is a respectable answer if you’re skiing with friends you actually like. Or if you’re just a magnanimous person leaving the lane open for those needing to squeeze in one more lap before work. (Bless you.)

But for the rest of us, the non-trivial solution is *maybe*, and that’s where it gets complicated. If you’ve stood in the singles line long enough, conducting a one-sided strategy session with yourself, you already know what the answer depends on: how long the lines are, how many lanes are feeding the lift, what sizes of groups are showing up, and how aggressively the liftie plays Tetris with strangers. It’s probability theory—sequential probabilities in an effectively infinite sequence—with human behavior complicating things.

Let’s fix ideas with a simple case: a quad chair (after the Latin *quattuor*, which, duh, means four). Suppose groups of two, three, and four arrive with equal probability. (God bless the skier who arrives at the head of the regular line solo and tows three singles up the mountain, but let’s assume him away.) The average group size is then three, which means there’s one empty seat per chair on average.

Now we can state a rule of thumb. If each chair has, on average, one open seat, then each chair takes one group and one single. The singles line moves as fast as the regular line, so you should take it when it’s shorter. If there are two regular lanes feeding the lift, take the singles line if it’s less than twice as long as either lane; if there are four, less than four times as long.

Based on empirical observation, that means the singles line is usually the move. At Gadzoom at Snowbird, for example, there are four lanes for regular traffic, but the singles line is never *four times* as long as any of those lanes.

Of course, that base case assumes groups arrive randomly and stay that way. But sometimes two groups of two reach the front together, and—either organically or at the direction of a well-meaning but catastrophically misguided liftie—merge into a four. The regular line moves faster and leaves fewer seats for singles!

In the simple case where groups of two, three, and four arrive with equal probability, those back-to-back pairs occur often enough that these merges matter: a group that starts 100th

in the regular line will reach the lift at about the same time as the 67th person in the singles line. (Like Fermat, I leave the proof to the reader.)

So the guidance tightens. With one regular lane, switch only if the singles line is roughly two-thirds as long as the regular line. With four lanes feeding the lift, switch only if it's less than about 2.7 times the length.

And that's still the polite version, because whether skiers tend to ride in groups of twos, threes, or fours is an empirical question—some hero of social science should stand at their local lift with a clipboard and count the actual distribution. (Volunteers will receive a letter of recommendation of dubious utility.) The mix matters greatly: the singles math deteriorates if there are more couples (more opportunities to merge) or more quads to begin with. If it's Valentine's Day and there are twice as many pairs as triples or fours, the average number of empty seats per chair drops to about 0.7. If there are twice as many fours instead, it's worse: only about 0.58 empty seats per chair. And because the same merges that eat empty seats also compress the regular line, the practical cutoff tightens toward three-fifths. Take the singles line only if it's less than about 60% of a single regular line on couple-heavy days, and closer to 55% when the mountain is mostly fours.

Chair size changes everything. Triples leave slightly less room for singles, even when quads split into doubles—about 0.75 seats per chair. Six-packs tell a more interesting story. They create new packing patterns— $2 + 2 + 2$, $3 + 3$, $4 + 2$, $5 + 1$ —but also much more total capacity. Run the simple “take the next group if it fits” model with equal odds of 2-, 3-, 4-, and 5-person groups, and the expected number of empty seats per chair is $414/384$ —about 1.08—so a bit better than your odds on a quad. If true sixes are common, however, the math swings against you, because a real six is a chair singles never touch. A singles line on a double chair? I've seen it, but I think it was there to meet cool people.

Once you decide to understand the singles line, you can spend your ride replaying edge cases: merging habits, the bureaucratic lane geometry of corporate resorts, the stoke when the liftie calls out four singles. It's amazing what you can solve while waiting for skiing.

So, should you take the singles line?

Maybe. Often. But not always. Take it if it's clearly shorter, if chairs are leaving with empty seats, and if the liftie is favorably disposed towards keeping the line moving. Not when you're with friends. Hopefully not on Valentine's Day. Accept that it's a gamble, but win or lose or draw, at least you're getting another lap.